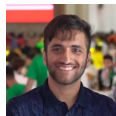


Value-iteration with Guessing for Markov chains and Markov decisions processes

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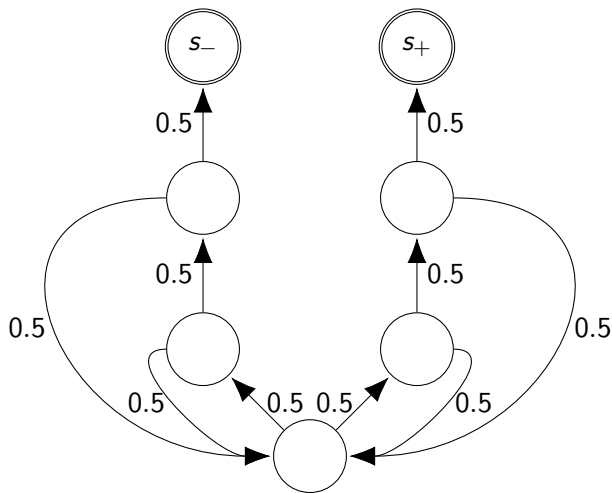
Jakub Svoboda



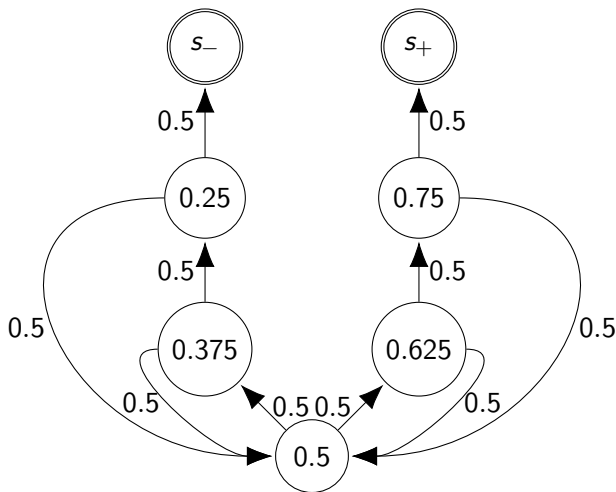
GAIMSS'24, Metz, France



Markov Chain



Reachability values



Update operator

For $v \in \mathbb{R}^S$,

$$\text{Update}(v)(s) := \max_{a \in A} \left\{ \sum_{s' \in S} p(s' | s, a) v(s') \right\}$$

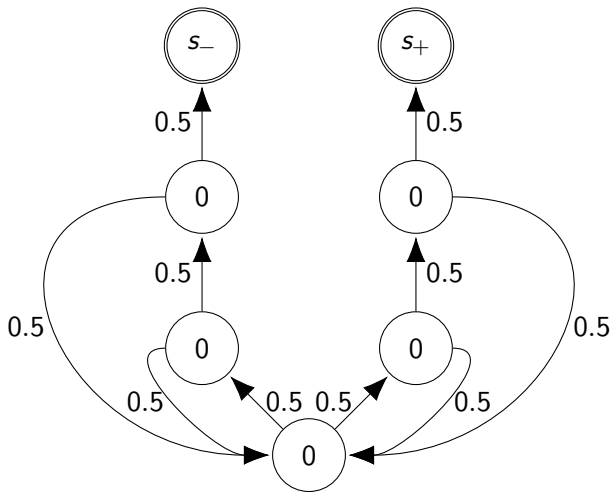
The reachability value is the least solution with

$$\text{val}(s_+) = 1 \qquad \text{val} = \text{Update}(\text{val})$$

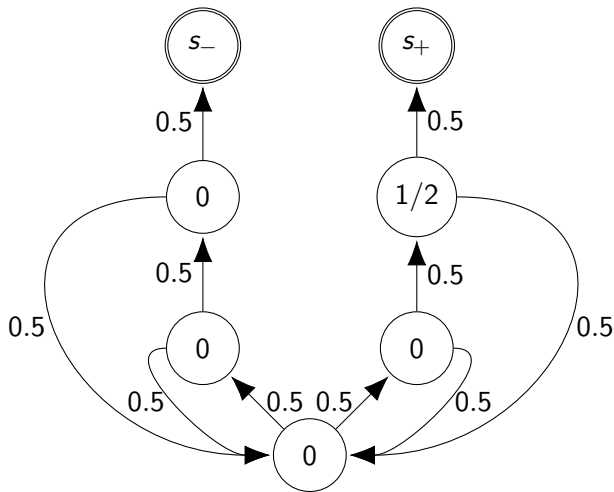
Moreover, if $v_0(s) = \mathbb{1}[s = s_+]$ and $v_{i+1} := \text{Update}(v_i)$, then

$$(v_i) \nearrow \text{val}$$

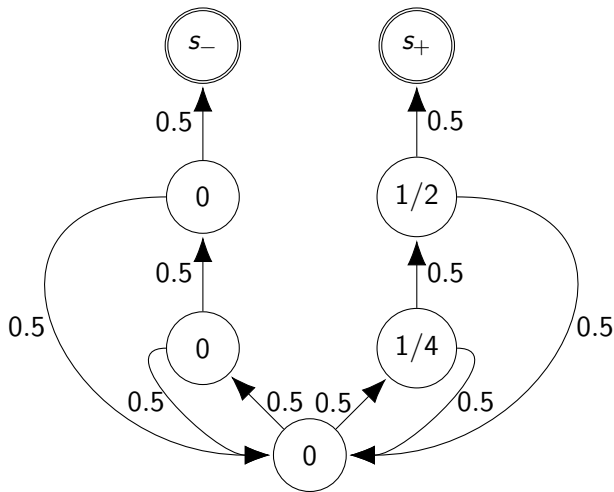
Value Iteration



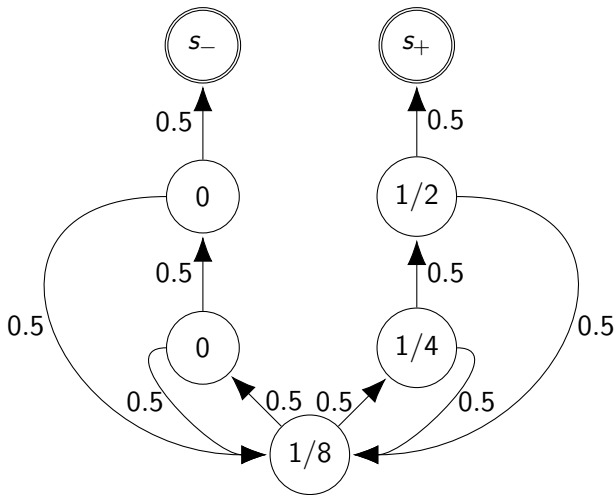
Value Iteration



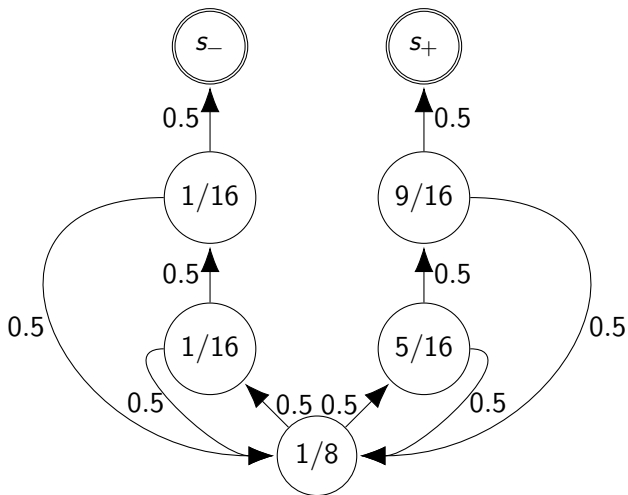
Value Iteration



Value Iteration



Value Iteration



Value Iteration speed

Lemma

The speed of convergence is driven by

$$\|v_{|S|i} - \text{val}\|_{\infty} \leq \left(1 - p_{\min}^{|S|}\right)^i \|v_0 - \text{val}\|_{\infty}$$

So, to have an ε -approximation of val, Value Iteration requires many iterations:

$$(-\log(\varepsilon)|S|/p_{\min})^{\Omega(|S|)}$$

*In other words, **exponentially** many.*

Value Iteration speed, improved

Lemma

To have an ε -approximation of val , we require fewer iterations:

$$(-\log(\varepsilon)|S|/p_{\min})^{\mathcal{O}(\sqrt{|S|})}$$

*In other words, **subexponentially** many.*